Context-Aware Background Music Descriptor Generation from Movie Scripts

Harshal Karangale[1] **Suyash Pande**[1] **Chaitanya Kadupukutla**[1] **Shivam Patel**[1]
[1]International Institute of Information Technology, Hyderabad
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Abstract

Film music plays a critical role in conveying emotional and narrative information in visual storytelling. Existing systems rely on coarse sentiment classification which fails to capture the deeper narrative context required for cinematic music generation. This paper presents **CineEmotion**, a hierarchical deep learning framework that converts screenplay dialogue into structured music descriptors by modelling emotional, scene-level, and narrative dynamics. We establish a baseline for scene-level emotion classification using two architectures trained on GoEmotions and domain-adapted to movie scripts: a BERT-based transformer with stacked encoder layers (**SELD**) and a frozen BERT feature extractor with a bidirectional LSTM (**BiLSTM-Attn**). Building on baseline findings, we propose a four-module pipeline comprising a Dialogue Transformer Encoder, Scene Encoder, Narrative LSTM, and Music Planner that predicts music attributes such as tempo, tonality, orchestration, and harmonic style. Experiments demonstrate the feasibility of mapping narrative context to structured musical descriptors using deep neural architectures.

1 Introduction

Music in cinema functions not only as an emotional amplifier but as a narrative device that conveys pacing, tension, and dramatic subtext. While recent advances in machine learning have enabled automatic music generation, most existing systems rely on surface-level sentiment or binary emotion labels. Such approaches are insufficient for film scoring, where narrative context plays a central role.

For example, a tense interpersonal argument and a fast-paced action sequence may both be classified as "intense," yet they demand entirely different musical styles. A character sitting in anxious silence in an elevator — as in the iconic elevator scene from *500 Days of Summer* where Tom musters the courage to speak to Summer only to ask "So how was your weekend?" — carries a fundamentally different emotional signature than an action confrontation, even if both register similar intensity scores under coarse sentiment systems.

This limitation motivates our work. We propose **CineEmotion**, a hierarchical pipeline that (1) performs multi-label emotion classification at the scene level, (2) maps predicted emotions to four cinematically meaningful emotional cores, and (3) infers the emotional arc of a full movie script to support downstream music descriptor generation.

Our contributions in this mid-submission are:

- A custom BERT-based transformer architecture (SELD) achieving macro F1 of 0.682 on GoEmotions with per-class threshold tuning.
- An automated annotation pipeline using the Gemini API to label 358 movie script scenes across three films with structured cinematic attributes (34 fields per scene).
- A domain adaptation procedure transferring GoEmotions-trained weights to movie script data, improving macro F1 from 0.18 to 0.45.
- A comparative evaluation of SELD and BiLSTM-Attn across direct inference and domain-adapted settings, quantifying the Reddit-to-script domain gap.
- A proposed four-module pipeline (Dialogue Transformer Encoder $\rightarrow$ Scene Encoder $\rightarrow$ Narrative LSTM $\rightarrow$ Music Planner) motivated by baseline findings, with preliminary results on SuperEmotion and MELD datasets.

2 Motivation and Objectives

Film composers typically interpret scripts by analysing emotional transitions, character interactions, and narrative tension. Automating parts of this process could assist composers and enable intelligent soundtrack generation systems.

Motivation

The main motivations of this project are:

- Develop a system that understands emotional and narrative structure from dialogue.
- Map narrative context to structured musical descriptors.
- Explore hierarchical representation learning across multiple temporal scales.
- Build a modular pipeline that integrates NLP and music representation.

Objectives

The primary objectives are:

1. Build a dialogue-level emotion encoder to extract emotional representations from utterances.
2. Model scene-level emotional dynamics using sequence models.
3. Capture narrative arcs across scenes using causal sequence modelling.
4. Generate structured music descriptors using a multi-task neural network.
5. Evaluate how effectively narrative features predict musical attributes.

3 Related Work

Fine-Grained Emotion Recognition. Early emotion analysis relied on coarse sentiment polarity. Demszky et al. (2020) introduced GoEmotions, a large-scale dataset of 58k Reddit comments annotated with 27 emotion categories plus neutral. This established that text emotions are multi-label, overlapping, and highly contextual, motivating richer representations beyond binary sentiment classification.

Context-Dependent Emotion Modeling. Poria et al. (2017) demonstrated that utterance-level emotion cannot be accurately interpreted in isolation. Using LSTM-based architectures incorporating surrounding utterances in video, they showed significant gains in emotion recognition, establishing the importance of dialogue history. This directly motivates our scene-level context modeling where action lines and multi-character dialogue must be interpreted together.

Hierarchical Context for Text Classification. Song et al. (2018) proposed a Context-LSTM-CNN architecture incorporating document-level context into sentence classification, demonstrating improved performance in emotion recognition tasks where contextual information is critical. This supports the use of hierarchical context modeling in long-form script understanding.

Long-Context Transformers. Chen et al. (2024) introduced Core Context Aware Transformers combining global context pooling with locality-preserving attention for efficient long-document modeling. While focused on language modeling efficiency, this highlights the importance of hierarchical context for narrative understanding, relevant to our scene-level arc inference.

Emotion-Driven Music Recommendation. Lee et al. (2020) proposed a music recommendation system for podcasts by detecting emotions from scripts, training multiple neural models on an emotion-labelled Twitter dataset and applying domain adaptation to podcast scripts. This directly motivates our two-stage approach: train on a general emotion corpus, then adapt to the target domain.

4 Problem Definition

Given a raw movie script $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$ comprising N scenes, where each scene s_i contains a scene header, action lines, and character dialogue, the objectives are:

1. **Multi-label Emotion Prediction**

   For each scene s_i in a film, the model predicts a multi-label emotion vector

   $$\hat{y}_i \in \{0, 1\}^K$$

   where $K = 9$ represents the number of emotion classes and $\hat{y}_i$ indicates the presence or absence of each emotion in scene i.

2. **Cinematic Emotional Core Mapping**

   The predicted emotion vector is mapped to a higher-level cinematic emotional category

   $$c_i \in \mathcal{C}$$

   where the set of cinematic emotional cores is defined as

$$\mathcal{C} = \{\textbf{Positive Uplifting}, \textbf{Neutral Complex}, \textbf{Tension Action}, \textbf{Negative Distressing}\}$$

3. **Emotional Arc Construction**

   The sequence of cinematic emotional cores across all scenes forms the emotional arc of the film

$$\mathcal{A} = \{c_1, c_2, \ldots, c_N\}$$

   where N is the total number of scenes. This sequence represents the narrative emotional trajectory of the film.

This mapping reduces fine-grained emotions into broader narrative emotional states.

5 Dataset

5.1 GoEmotions

GoEmotions (Demszky et al., 2020) contains 58,009 English Reddit comments annotated with one or more of 27 emotion categories plus neutral. We filter to 9 labels with training set frequency exceeding 2,000 examples to focus on well-represented classes and reduce label sparsity. Table 1 shows the selected labels with their cinematic core mappings.

Emotion	Train Count	Core
neutral	14,219	Neutral_Complex
admiration	4,130	Positive_Uplifting
approval	2,939	Positive_Uplifting
gratitude	2,662	Positive_Uplifting
annoyance	2,470	Negative_Distressing
disapproval	2,470	Negative_Distressing
amusement	2,328	Positive_Uplifting
curiosity	2,191	Tension_Action
love	2,086	Positive_Uplifting

Table 1: Selected GoEmotions labels with training counts and cinematic core mapping. Neutral dominates at 30% of samples.

The dataset is split into train (43,410), validation (5,426), and test (5,427) sets following the official split. The label distribution is highly imbalanced, which we address using label smoothing in training.

5.2 Movie Script Dataset

To evaluate real-world performance and perform domain adaptation, we annotated scenes from three films using the Gemini API:

- *500 Days of Summer* (romantic drama): 107 scenes
- *Avatar* (sci-fi action): 79 scenes

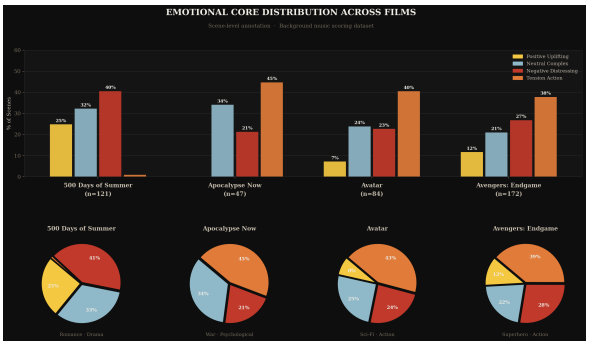

Figure 1: Emotional core distribution across all four films (bar chart and per-film pie charts). *500 Days of Summer* is dominated by `Negative_Distressing` (41%) reflecting its introspective drama genre. Action films (Avatar, Avengers) show substantially higher `Tension_Action` proportions (43% and 39% respectively), confirming genre-level validity of the Gemini annotations.

- *Avengers: Endgame* (superhero action): 172 scenes

Each scene was annotated with a rich structured schema including: `emotional_core` (one of 4 cinematic cores), `tension_level` (1–10), `pacing_intensity` (1–10), `narrative_stage`, `character_internal_state`, `conflict_nature`, `conflict_type`, `emotional_intensity`, `action_intensity`, `musical_cue_type`, `foreshadowing_callback`, `relationship_status`, and `thematic_elements`. The pipeline also produces `validation_issues` flags for out-of-range or inconsistent values.

Table 2 shows the distribution across films.

Film	Total	PU	NC	TA	ND
500 Days	107	27	35	1	44
Avatar	79	6	20	34	19
Avengers	172	20	36	65	46
Total	**358**	53	91	100	109

Table 2: Scene distribution per cinematic core across films. PU=Positive_Uplifting, NC=Neutral_Complex, TA=Tension_Action, ND=Negative_Distressing.

500 Days of Summer is dominated by `Negative_Distressing` (41%) reflecting its introspective genre, while Avatar and Avengers contribute the majority of `Tension_Action` scenes. This diversity across genres is essential for balanced domain adaptation Figure 1.

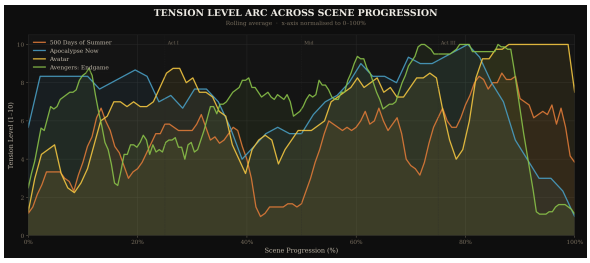

Figure 2: Tension level arc across normalised scene progression (0–100%) for all four films, computed as a rolling average over annotated `tension_level` values (1–10). Act boundaries are marked at roughly 33% and 66%. All action films (*Apocalypse Now*, *Avatar*, *Avengers: Endgame*) sustain tension above 6 throughout Act II–III. *500 Days of Summer* exhibits a distinctive mid-film tension trough consistent with its romantic drama structure.

5.3 Schema Induction

The annotation schema was not designed by hand. It was induced from real screenplays through a three-stage automated process (Figure 4). Table 19 shows a complete annotation schema.

Stage 1 — Per-Script Field Discovery. Eleven complete screenplays spanning eight or more genres were each passed in full to Gemini 2.5 Flash (1M token context). The model was prompted as a film music supervisor reading scene-by-scene, maintaining running memory of character states, tensions, and narrative events. For each script it was asked: *"What does a composer need to know about this scene to write the right music — given everything that has happened in the story so far?"* It named its own fields, defined their value sets, and rated their scoring impact with no preset list given. This produced 11 per-script schemas of 21–27 fields each.

Stage 2 — Merge. All 11 schemas were passed to a second Gemini call acting as a cross-genre music supervisor. It unified semantically duplicate fields (e.g., two scripts' "tension type" variants collapsed into one), took the union of all values per field, capped categorical fields at 30 values through generalisation, and dropped fields appearing in only one script as too scene-specific to generalise. This produced approximately 40 candidate fields.

Stage 3 — Manual Curation. The 40 fields were reviewed manually: names standardised to `snake_case`, remaining semantic overlaps resolved, and fields assigned to three tiers (Essential / High Impact / Contextual), fixing the final count

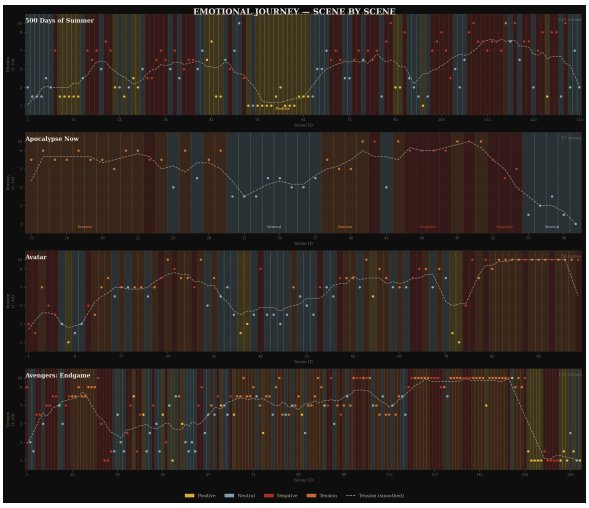

Figure 3: Scene-by-scene emotional journey for all four films. Coloured vertical bars indicate the dominant `emotional_core` per scene; overlaid dots show per-scene `tension_level`; the dashed white line is the smoothed tension trajectory. The plot reveals narrative rhythm: *500 Days of Summer* shows frequent alternation between positive and negative cores reflecting the non-linear timeline, while *Avengers: Endgame* shows a sustained high-tension third act.

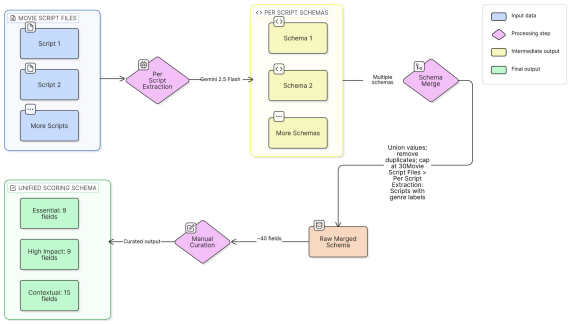

Figure 4: Schema extraction

at 33. A 34th field — `internal_cue_points` — was added as a shot-level layer marking intra-scene music shift moments.

The key constraint throughout: every field must be reusable across all scene types and genres and must directly drive a music decision (tempo, instrumentation, emotional palette, harmonic language, silence, leitmotif activation). Fields applying to only one film or moment were rejected. Table 19 summarises all 34 fields grouped by function.

5.4 Annotation Pipeline

Each screenplay is split into individual scenes using `INT./EXT.` scene headers as delimiters and processed in order through the Gemini API. The model is explicitly instructed to use the accumulated story

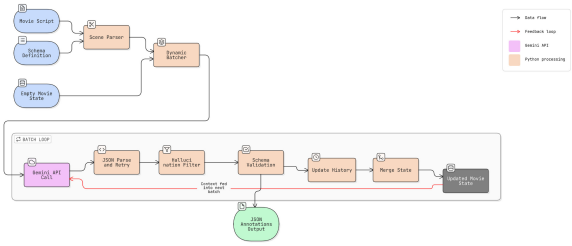

Figure 5: Dataset Annotation Pipeline Architecture

state when assigning every field value — not to read each scene in isolation. Because annotating all scenes at once would exceed the API output token limit (8,192 tokens), scenes are processed in batches:

$$\text{scenes per batch} = \lfloor 7500 \div 600 \rfloor \approx 12 \quad (1)$$

After each batch, Gemini returns both annotations and an `updated_movie_state` — a compact summary organised into seven buckets (narrative, tension, character, relationship, motif, moral, memory state) — passed as input to the next batch (Figure 5). History lists (`emotional_arc`, `key_events`, `trajectory_history`) are built entirely in Python by diffing consecutive states, so Gemini never outputs history directly, saving tokens while preserving complete character arcs. A safe merge function prevents LLM forgetting: lists only grow, scalars take the new value, and `peak_tension_so_far` can only increase. Every annotation is validated on return and confidence-penalised for missing required fields or hallucinated character names.The `movie_state` stores scalars only (current values).

A safe merge function protects against LLM forgetting. Rather than replacing the old state with the new state returned by Gemini, the pipeline merges them: lists only grow (established motifs and characters cannot vanish), scalars take the new value, and `peak_tension_so_far` can only increase. This prevents a single batch of poor model output from corrupting the accumulated story context.

Every annotation is validated on return: numeric fields must fall within their defined range, categorical fields must belong to the allowed value set, and missing required fields are flagged and penalised in the confidence score. Character names in `character_states` that do not appear in the batch scene text are detected as hallucinations and removed before merging. The final output per script is a single JSON containing all scene annotations, each with its 34 field values, confidence score, validation issues, and the original scene text alongside the labels

Figure 6 shows a complete annotation for scene 10 of *Avengers: Endgame*, illustrating how the accumulated context state from prior batches directly informs the annotation of the current scene.

5.5 SuperEmotion and MELD Datasets

The proposed four-module pipeline (Section 8) is trained on two additional datasets targeting dialogue-level emotion recognition.

SuperEmotion. SuperEmotion is a large-scale aggregated corpus combining multiple emotion-labelled dialogue datasets into a unified 7-class label space (`anger`, `disgust`, `fear`, `joy`, `neutral`, `sadness`, `surprise`). Module 1 (Dialogue Transformer Encoder) is pre-trained on SuperEmotion for 12 epochs using plain cross-entropy loss with label smoothing ($\varepsilon = 0.05$), achieving a best validation F1 of 75.58%. The large scale and label diversity of SuperEmotion provides a strong initialisation for dialogue-level emotion representations before fine-tuning on conversational data.

MELD. The Multimodal EmotionLines Dataset (Poria et al., 2019) contains 1,433 dialogues and 13,708 utterances extracted from the *Friends* TV series, annotated with the same 7 emotion categories. MELD is used for Phase 2 fine-tuning of Module 1, with a `NeutralPenaltyLoss` applying a neutral class weight of $\times 1.8$ to counteract the strong neutral dominance observed in conversational data — directly addressing the neutral collapse failure mode identified in our baseline experiments (Section 7).

6 Baseline: Scene-Level Emotion Classification

We first establish a baseline for scene-level emotion classification on movie scripts using two architectures. The findings from this baseline — particularly the domain gap, neutral collapse, and data bottleneck — directly motivate each design decision in the proposed full pipeline (Section 8). We evaluate two architectures: The first, **SELD** (Scene-Level Emotion with Learned Dynamics), on BERT embeddings end-to-end with stacked custom Transformer encoder layers. The second, **BiLSTM-Attn**, keeps BERT frozen and trains a bidirectional

INPUT — Scene 10

`INT. BENATAR, FLIGHT DECK – NIGHT`

Tony stands on the flight deck, one hand resting on the glass, staring into the unknown.

TONY
"Pep, I know I said no more surprises. But, I gotta say, I was really hoping to pull off one last one. But it looks like… well, you know what it looks like."

Context state passed in:
narrative_stage: Middle_Development
tension trajectory: escalating (peak: 8)
Tony emotional arc:
Despairing → Resigned
active foreshadowing:
"Tony adrift, dying"

Gemini +ctx →

OUTPUT — Annotation confidence: **0.97**

Narrative & Scale	
narrative_stage	Middle_Development
sense_of_scale	Intimate_Personal
Emotion & Tone	
emotional_core	**Negative_Distressing**
tension / intensity / pacing	8 / 9 / 4
dramatic_shift	**true**
humor_tone	None moral_ambiguity: 2
Conflict	
conflict_nature	Psychological_Tension
conflict_type	Internal_Personal
action / violence	1 / 0
Character	
character_focus	Protagonist_Individual
internal_state	**Resigned**
relationship	Established_Intimate
Narrative Device	
foreshadowing	Impending_Danger_Loss
thematic	Personal_Journey, Relationships
Cinematic Style	
musical_cue	Original_Score_Underscore
dialogue	High_Dominant
visual_pacing	Slow_Pensive
soundscape	Mechanical, Human_Vocal
cue_point	*Key_Line: "one last one"*
World & Setting	
scene_setting	Interior_Confined
time_period	Future_SciFi
tech / spiritual	7 / 1 mystical: None

Figure 6: Two-panel annotation example, scene 10 of *Avengers: Endgame*. **Left (blue):** raw scene text and the `movie_state` context accumulated from prior batches, showing Tony's emotional arc (*Despairing→Resigned*) and the active foreshadowing thread. **Right (green):** the 34-field annotation produced using both scene text and context. `emotional_core=Negative_Distressing` follows from the sustained escalating trajectory; `internal_state=Resigned` follows from the arc; `foreshadowing= Impending_Danger_Loss` from the unresolved thread. Confidence 0.97.

LSTM with attention on top of fixed BERT representations. Both are first trained on GoEmotions for 9-class multi-label emotion classification, then domain-adapted to 4-class cinematic core prediction on annotated movie script scenes.

6.1 SELD: Scene-Level Emotion with Learned Dynamics

6.1.1 Architecture Overview

The forward pass proceeds as follows:

1. **BERT Encoder:** Input tokens are passed through `bert-base-uncased` (12 layers, 768 hidden dimensions). Only the contextual token embeddings (`last_hidden_state`) $\mathbf{H} \in \mathbb{R}^{B \times T \times 768}$ are used, where $T = 128$. BERT weights are kept frozen throughout training; only the custom encoder layers, attention pooling, and classifier head are updated.
2. **Custom Transformer Encoder Layers:** Two stacked `TransformerEncoderLayer` blocks applied on top of BERT's output, each with multi-head self-attention (8 heads), feed-forward network (hidden dimension 2048), layer normalisation, and residual connections with dropout ($p = 0.2$).
3. **Attention Pooling:** A learned scorer collapses the sequence into a single vector.
4. **Classifier Head:** A two-layer MLP maps the pooled representation to class logits with sigmoid activation.

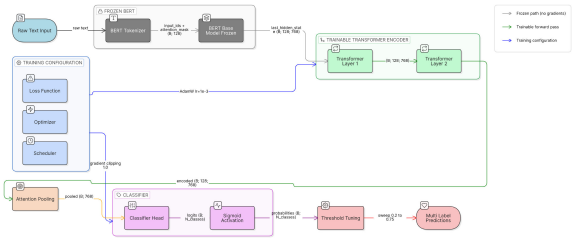

Figure 7: Transformer Architecture

6.1.2 Multi-Head Self-Attention

Each custom encoder layer uses scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (2)$$

where $d_k = 768/8 = 96$. The tokeniser attention mask prevents attending to padding positions:

$$\tilde{s}_{ij} = \begin{cases} s_{ij} & \text{if } m_j = 1 \\ -\infty & \text{if } m_j = 0 \end{cases} \quad (3)$$

Each layer applies pre-norm residual connections:

$$\mathbf{x}' = \text{LayerNorm}(\mathbf{x} + \text{MHSA}(\mathbf{x})) \quad (4)$$

$$\mathbf{x}'' = \text{LayerNorm}(\mathbf{x}' + \text{FFN}(\mathbf{x}')) \quad (5)$$

6.1.3 Attention Pooling

A learned scorer assigns scalar importance to each token:

$$e_t = \mathbf{w}_2^\top \tanh(\mathbf{W}_1 \mathbf{h}_t + \mathbf{b}_1) \quad (6)$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})}, \qquad \mathbf{z} = \sum_t \alpha_t \mathbf{h}_t \quad (7)$$

where $\mathbf{W}_1 \in \mathbb{R}^{256\times768}$, $\mathbf{w}_2 \in \mathbb{R}^{256}$, and padding is masked before softmax. This focuses the model on emotionally salient dialogue tokens.

6.1.4 Classifier Head

$$\hat{y} = \sigma(\mathbf{W}_2\, \text{GELU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + \mathbf{b}_2) \quad (8)$$

where $\mathbf{W}_1 \in \mathbb{R}^{512\times768}$, $\mathbf{W}_2 \in \mathbb{R}^{9\times512}$, dropout $p = 0.2$. Sigmoid is used for multi-label output.

6.1.5 Loss and Training

We use Label Smoothing BCE to prevent overconfidence on the dominant neutral class ($\varepsilon = 0.05$):

$$\tilde{y}_k = y_k(1-\varepsilon) + \tfrac{\varepsilon}{2} \quad (9)$$

$$\mathcal{L} = -\frac{1}{K}\sum_k \left[\tilde{y}_k \log \hat{y}_k + (1-\tilde{y}_k)\log(1-\hat{y}_k)\right] \quad (10)$$

Hyperparameter	Value
MAX_LEN	128 tokens
Batch size	32
Epochs	6
BERT LR	2×10^{-5}
Custom layer LR	1×10^{-4}
LR schedule	Cosine with 10% warmup
Optimiser	AdamW ($\lambda = 0.01$)
Gradient clip	1.0
Label smoothing	$\varepsilon = 0.05$
FF dimension	2048
Attention heads	8
Dropout	0.2

Table 3: SELD training hyperparameters.

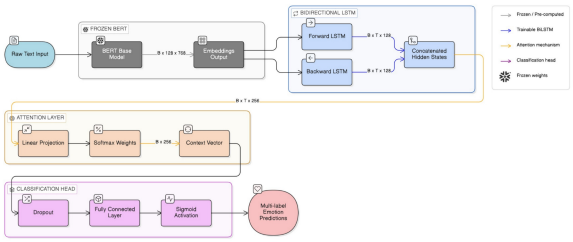

Figure 8: BiLSTM Architecture

6.1.6 Per-Class Threshold Tuning

After training, a per-class decision threshold τ_k is tuned on the validation set:

$$\tau_k^* = \arg\max_{\tau\in[0.2,0.75]} \text{F1}(\mathbf{y}_k, \mathbf{1}[\hat{p}_k > \tau]) \quad (11)$$

This yields a final test macro F1 of **0.682**.

6.2 BiLSTM-Attn: Frozen BERT Feature Extractor

The second architecture keeps BERT frozen throughout GoEmotions training — only the LSTM, attention, and classifier parameters are learned.

6.2.1 Bidirectional LSTM

$$\overrightarrow{\mathbf{h}}_t = \text{LSTM}_{\rightarrow}(\mathbf{h}_t^{\text{BERT}}, \overrightarrow{\mathbf{h}}_{t-1}) \quad (12)$$

$$\overleftarrow{\mathbf{h}}_t = \text{LSTM}_{\leftarrow}(\mathbf{h}_t^{\text{BERT}}, \overleftarrow{\mathbf{h}}_{t+1}) \quad (13)$$

$$\mathbf{o}_t = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{256} \quad (14)$$

Hidden size 128 per direction (256 bidirectional), 2 stacked layers.

6.2.2 Attention and Classifier

$$\alpha_t = \text{softmax}(\mathbf{w}^\top \mathbf{o}_t), \qquad \mathbf{c} = \sum_t \alpha_t\, \mathbf{o}_t \quad (15)$$

$$\hat{y} = \sigma(\mathbf{W}_c\, \text{Dropout}_{0.3}(\mathbf{c}) + \mathbf{b}_c) \quad (16)$$

where $\mathbf{W}_c \in \mathbb{R}^{9\times256}$.

Hyperparameter	Value
BERT backbone	`bert-base-uncased` (frozen)
LSTM hidden size	128 per direction (256 bidir.)
LSTM layers	2
MAX_LEN	64 tokens
Batch size	16
Dropout	0.3
Optimiser	Adam, lr $= 2 \times 10^{-4}$
Loss	BCEWithLogitsLoss

Table 4: BiLSTM-Attn training hyperparameters.

Property	SELD	BiLSTM-Attn
Sequence model	Transformer	BiLSTM
Pooling	Attn (768-dim)	Attn (256-dim)
Classifier	768→512→9	256→9
MAX_LEN	128 tokens	64 tokens
Trainable params	∼120M	∼6M

Table 5: Architectural comparison. LS-BCE = Label Smoothing BCE.

6.2.3 Architectural Comparison

6.3 Emotion-to-Cinematic-Core Mapping

We define a deterministic mapping $\phi : \mathcal{E} \to \mathcal{C}$ from 9 GoEmotions labels to 4 cinematic cores. The predicted core is determined by max-pooling:

$$\text{score}(c) = \max_{\{e:\phi(e)=c\}} \hat{p}_e, \quad \hat{c} = \arg\max_{c\in\mathcal{C}} \text{score}(c) \tag{17}$$

The Avengers dataset introduced extra Gemini labels resolved via an extended mapping (e.g., `Awe_Wonder`→Positive_Uplifting, `Dread_Fear`→Negative_Distressing, `Suspense_Unease`→Tension_Action).

6.4 Domain Adaptation

GoEmotions Reddit comments average 15–20 tokens; movie scenes average 60–120 words with multi-character dialogue and narrative subtext. This domain shift causes both models to collapse to `Neutral_Complex` without adaptation.

6.4.1 Dataset and Split

The balanced test set uses 11 scenes per class (44 total):

$$n_{\text{test}} = \max\Big(5,\ \lfloor 0.2 \times \min_c |\mathcal{D}_c| \rfloor\Big) = \lfloor 0.2{\times}58 \rfloor = 11 \tag{18}$$

Class weights $w_c = N_{\text{train}}/(|\mathcal{C}| \cdot n_c^{\text{train}})$ are applied in CrossEntropyLoss.

GoEmotions Label	Cinematic Core
neutral	Neutral_Complex
admiration, approval, amusement	Positive_Uplifting
gratitude, love	Positive_Uplifting
curiosity	Tension_Action
annoyance, disapproval	Negative_Distressing

Table 6: GoEmotions to cinematic core mapping.

Config	SELD+DA	BiLSTM+DA
Total params	120.9M	110.8M
Trainable params	40.4M (33.4%)	30.3M (27.3%)
Frozen BERT layers	0–7	0–7
Trainable BERT	8–11	8–11
BERT LR	10^{-5}	10^{-5}
Custom LR	5×10^{-5}	5×10^{-5}
Optimiser	AdamW	AdamW
LR schedule	Cosine+warmup	Cosine+warmup
Batch size	8	8
Max epochs	40	40
Early stopping	Patience 7	Patience 7
Train / Test	286 / 72	314 / 44
PU weight	1.55	1.67
NC weight	0.98	0.98
TA weight	0.89	0.88
ND weight	0.82	0.80

Table 7: Domain adaptation configuration for both architectures.

6.4.2 SELD Domain Adaptation

The 9-class head is replaced with a 4-class softmax head (768→256→4). BERT embeddings and bottom 8 encoder layers are frozen; top 4 encoder layers and all custom layers are trainable:

$$\theta_{\text{trainable}} = \theta_{\text{BERT}}^{[8:12]} \cup \theta_{\text{enc1,2}} \cup \theta_{\text{pool,head}}$$

A stratified 80/20 split across all three films yields 286 training and 72 test scenes. Class weights are applied in CrossEntropyLoss to address label imbalance, and the adapted classifier head uses dropout $p = 0.4$.

7 Experiments and Results

7.1 GoEmotions Performance

7.2 Direct Inference on Movie Scripts

Both models are tested directly on 107 labelled scenes from *500 Days of Summer* without adaptation.

Three failure modes are consistent across both architectures: (1) **Negative_Distressing collapse**: movie script distress is expressed through silence and subtext, not explicit negative language; (2) **Tension_Action failure** (F1=0.00): only 1 such scene exists in a romantic drama; (3) **Neutral_Complex**

Emotion	SELD F1
gratitude	0.922
amusement	0.854
love	0.810
neutral	0.755
admiration	0.738
curiosity	0.658
disapproval	0.478
annoyance	0.475
approval	0.448
Macro F1	**0.682**

Table 8: SELD per-class F1 on GoEmotions test set (tuned thresholds). BiLSTM-Attn did not exceed 0.40 macro F1, attributed to frozen BERT representations.

Core	Prec.	Rec.	F1	Sup.
SELD (Macro F1 = 0.180)				
Positive_Uplifting	0.21	0.26	0.23	27
Neutral_Complex	0.29	0.51	0.37	35
Tension_Action	0.00	0.00	0.00	1
Negative_Distressing	0.43	0.07	0.12	44
Macro avg	0.23	0.21	0.18	107
BiLSTM-Attn (Macro F1 = 0.20)				
Positive_Uplifting	0.27	0.26	0.26	27
Neutral_Complex	0.34	0.71	0.46	35
Tension_Action	0.00	0.00	0.00	1
Negative_Distressing	0.50	0.05	0.08	44
Macro avg	0.28	0.26	0.20	107

Table 9: Direct inference on *500 Days of Summer* (107 scenes). Both models fail on Negative_Distressing and Tension_Action.

over-prediction: GoEmotions neutral bias (14,219 samples, 30%) transfers to the movie domain. The domain gap represents a 67.7% relative macro F1 drop (0.682→0.22).

7.3 After Domain Adaptation

Ep.	Tr.L	Tr.Acc	Te.L	Te.Acc	F1	Gap
SELD DA (best: epoch 30, Train=286, Test=72)						
1	1.385	0.224	1.386	0.222	0.179	0.002
5	1.345	0.413	1.367	0.319	0.312	0.093
10	1.196	0.542	1.347	0.347	0.330	0.195
30	**0.653**	**0.727**	**1.588**	**0.458**	**0.450**	**0.269**
37	0.625	0.759	1.649	0.431	0.422	0.328
BiLSTM DA (best: epoch 15, Train=314, Test=44)						
1	1.450	0.268	1.448	0.227	0.141	0.040
5	1.306	0.404	1.423	0.295	0.247	0.109
10	1.042	0.624	1.342	0.386	0.376	0.238
15	**0.731**	**0.828**	**1.364**	**0.409**	**0.403**	**0.419**
22	0.431	0.924	1.500	0.341	0.334	0.583

Table 11: Training dynamics for both DA models (selected epochs). Both exhibit widening train-test gaps indicating overfitting from small training sets. SELD early-stops at epoch 11; BiLSTM at epoch 22.

Core	GT	GT%	SELD	BiLSTM
Positive_Uplifting	27	25.2%	34	26
Neutral_Complex	35	32.7%	62	74
Tension_Action	1	0.9%	4	3
Negative_Distressing	44	41.1%	7	4

Table 10: GT vs predicted distribution (no adaptation). Both models over-predict Neutral_Complex and miss Negative_Distressing (recall ≈0.05–0.07).

Core	Prec.	Rec.	F1	Sup.
SELD+DA (Macro F1=0.450, Test=72)				
Positive_Uplifting	0.33	0.50	0.40	12
Neutral_Complex	0.43	0.33	0.38	18
Tension_Action	0.52	0.55	0.54	20
Negative_Distressing	0.53	0.45	0.49	22
Macro avg	0.45	0.46	0.45	72
BiLSTM+DA (Macro F1=0.403, Test=44 balanced)				
Positive_Uplifting	0.30	0.27	0.29	11
Neutral_Complex	0.38	0.45	0.42	11
Tension_Action	0.58	0.64	0.61	11
Negative_Distressing	0.33	0.27	0.30	11
Macro avg	0.40	0.41	0.40	44

Table 12: Classification reports after domain adaptation.

7.4 Full Comparison

Model	Setting	Mac.F1	PU	NC	TA	ND
SELD	GoEmotions	0.682	–	–	–	–
BiLSTM	GoEmotions	<0.40	–	–	–	–
SELD	Direct	0.180	0.23	0.37	0.00	0.12
BiLSTM	Direct	0.200	0.26	0.46	0.00	0.08
SELD+DA	Adapted	**0.450**	**0.40**	0.38	0.54	**0.49**
BiLSTM+DA	Adapted	0.403	0.29	**0.42**	**0.61**	0.30

Table 13: Consolidated results. PU=Positive_Uplifting, NC=Neutral_Complex, TA=Tension_Action, ND=Negative_Distressing. Domain adaptation improves SELD by 150% (0.180→0.450) and BiLSTM by 101% (0.200→0.403).

Domain adaptation consistently improves both models. `Tension_Action` shows the largest gains, driven by Avatar and Avengers contributing 99 of 100 total TA scenes. `Negative_Distressing` recall recovers from ∼0.05 to 0.27–0.41. The persistent train-test gap (up to 0.58 in both models) indicates overfitting; we estimate a minimum of 500 labelled scenes per genre for stable generalisation.

8 Proposed Methodology and Architecture

The baseline results in Section 7 expose three fundamental limitations of treating script emo-

Module	Input	Core Layer	Output	Params
M1 DTE	tok(128)+pos(128)+spk(64)+turn(32)→**352d**	4× PreLN Transformer (d=256, 4h, ff=512) + GRU(256→128)	($U\times$**256d**) utt vecs	7.66M
M2 Scene	($U\times$256) + scene_type(**32d**)	2× BiLSTM (hidden=128→**256d**) + attn pool	**256d** scene vec + **7d** emo_probs	1.01M
M3 Narrative	**256d** scene vec	Causal LSTM(256→128) + TensionCell(1) + CharBank(**64d**) → fuse 193→**256d**	**256d** narr vec + τ(1)	546K
M4 Planner	256+256+64=**576d**	Trunk 576→512→256→**128d** (shared)	13 heads from **128d**	492K
			Total	**10.2M**

Table 14: All four modules with key dimensions, layer configurations, and parameter counts. Bold values highlight critical dimensionalities at each stage.

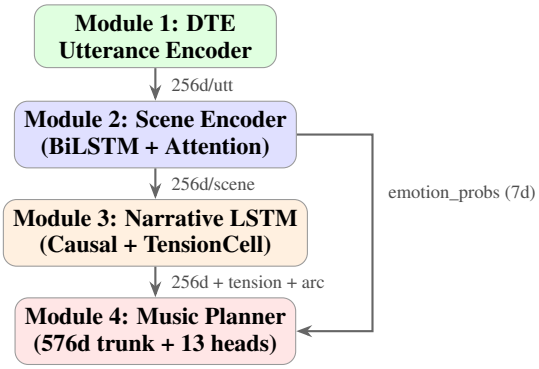

Figure 9: Proposed pipeline overview. M4 receives inputs from both M2 (emotion distribution) and M3 (narrative context). The baseline SELD/BiLSTM-Attn work corresponds to a compressed approximation of M1+M2, validating the scene-level emotion signal before full pipeline construction.

Type	Heads	Targets	Dims
Regression	3	tempo, intensity, valence	1, 1, 1
Classification	9	tonality, harmonic, motion, rhythm, density, dynamic, texture, transition, arc	3, 8, 6, 6, 3, 8, 6, 5, 5
Multi-label	1	instruments	15 (BCE)
Total	**13**		

Table 15: M4 output head summary. All 13 heads read from the shared 128d music state.

tion classification as a flat text classification problem: (1) the Reddit-to-script domain gap collapses predictions to `Neutral_Complex`; (2) scene-level text concatenation discards utterance-level speaker dynamics; and (3) no mechanism captures narrative arc progression across scenes. These findings motivate a four-module pipeline that processes a screenplay end-to-end from raw dialogue to structured music descriptors. Figure 9 shows the overall flow. Crucially, the four cinematic cores established in the baseline — `Positive_Uplifting`, `Neutral_Complex`, `Tension_Action`, `Negative_Distressing` — serve as the interface layer between the emotion classification modules and the downstream music planner.

8.1 Module 1: Dialogue Transformer Encoder

Input representation. Token embedding (128d) + positional embedding (128d) + speaker embedding (64d) + turn embedding (32d) → projected to 256d. This explicit speaker and turn modelling directly addresses the baseline limitation of treating multi-character dialogue as flat text.

Architecture. Four pre-LayerNorm Transformer blocks ($d_{\text{model}} = 256$, 4 heads, $d_{\text{ff}} = 512$). An *EmotionQueryToken* (EQT) encodes learnable class prototypes as a prepended CLS token. A *DialogueContextGRU* encodes the previous five utterances into a 128d context vector, fused with the CLS output → 256d final utterance representation. Total: 7.66M parameters.

Training. **Phase 1**: Pre-train on SuperEmotion (12 epochs, plain cross-entropy, label smoothing 0.05); best val F1 = 75.58%. **Phase 2**: Fine-tune on MELD (20 epochs, NeutralPenaltyLoss with neutral weight $\times$1.8, directly addressing the neutral collapse observed in the baseline); speaker embedding re-initialised from $|\text{vocab}| = 2$ to $|\text{vocab}| = 500$. Post-training threshold calibration raises test macro-F1 from 20.89% → **31.95%**.

8.2 Module 2: Scene Encoder

A 2-layer bidirectional LSTM (hidden=128, output=256d) processes the variable-length sequence of M1 utterance vectors — in contrast to the base-

line where raw scene text is passed directly to BERT, discarding utterance boundaries. Scene-type conditioning (32d embedding broadcast over the sequence) and masked attention pooling produce a single 256d scene vector. Eight scene-level and two per-utterance prediction heads are trained simultaneously with a multi-task loss.

All training labels are self-supervised: dominant emotion via L2-norm-weighted voting; scene type via heuristic rules; intensity via normalised L2 norm of DTE vectors; transition matrices from bigram counts. Class weights are clamped at 15.0 to prevent instability from the zero-sample *peaceful* class. Total: 1.0M parameters.

8.3 Module 3: Narrative LSTM

A forward-only (causal) 2-layer LSTM reads the episode's scene vector sequence, directly modelling the arc progression that the baseline's cinematic core sequence $\mathcal{A} = \{c_1, \ldots, c_N\}$ approximates. At each step t, three mechanisms are fused:

1. **Causal LSTM** $\rightarrow$ 128d hidden state.
2. **TensionMemoryCell**: three learned gates (γ decay, α arousal, ρ release) produce a scalar tension estimate:

$$\tau_t = \text{clip}(\gamma\, \tau_{t-1} + \alpha - \rho,\ 0, 1)$$

3. **CharacterMemoryBank**: a GRUCell maintains a 64d emotional state per character, updated on each appearance.

These are concatenated ($128 + 1 + 64 = 193$d) and projected to 256d narrative vectors. Arc phases are self-supervised via scene-position quintiles (Freytag's pyramid approximation). Total: 546K parameters.

8.4 Module 4: Music Planner

Input. Scene vector (256d) + narrative vector (256d) + emotion context (7d emotion probs $\rightarrow$ 64d via EmotionContextEncoder) = 576d.

Shared trunk. $576 \rightarrow 512 \rightarrow 256 \rightarrow 128$ (three bottleneck layers, LayerNorm, ReLU, dropout). The shared trunk forces internal consistency: all 13 heads read from the same 128d "music state", suppressing contradictory predictions (e.g. fast tempo + sustained rhythm).

13 output heads. Three regression targets (tempo 40–160 BPM, intensity $[0, 1]$, valence $[-1, 1]$); nine classification targets (tonality/3, harmonic style/8, motion type/6, rhythm style/6, density/3, dynamic shape/8, texture/6, transition/5, arc verify/5); one multi-label instrument head (15 groups, BCELoss). All targets are derived from M2+M3 signals via film-scoring rules: e.g. high arousal + tension $\rightarrow$ faster tempo; *sad* $\rightarrow$ melodic minor; *action* $\rightarrow$ chromatic aggressive; rising/climax arc $\rightarrow$ crescendo. Total: 492K parameters.

Module	Head / Task	F1	Acc.
M1	Emotion (argmax)	20.89%	—
	Emotion (calib.)	**31.95%**	—
M2	Dominant emotion	55.32%	70.15%
	(best val.)	55.63%	—
M3	Arc phase	52.47%	55.07%
	(best val.)	48.26%	—
M4	Harmonic style	74.14%	78.50%
	Dynamic shape	69.03%	85.00%
	Motion type	74.78%	83.50%
	Texture	70.62%	75.50%
	Tonality	86.26%	86.50%
	Transition	69.82%	92.00%
	Rhythm style	45.65%	85.50%
	Avg.	**70.04%**	—

Table 16: Proposed pipeline test performance (SuperEmotion/MELD). M4 average is over 7 classification heads. End-to-end evaluation on movie scripts is planned.

8.5 Proposed Pipeline Results

Table 16 summarises performance across all four modules on held-out test sets. Note that these results are obtained on the SuperEmotion and MELD datasets; evaluation on annotated movie scripts (the target domain established in Section 6.4) is planned as the next integration step.

Module 1's calibrated F1 was 31.95%. The gain from 31.95% (per-utterance, M1) to 55.32% (per-scene, M2) confirms that attention-pooled scene aggregation over utterance vectors substantially reduces label noise — the key architectural improvement over the baseline's flat scene concatenation. Module 3's 52.47% arc-phase F1 is notable given only 180 training episodes. Module 4's 70.04% average F1 demonstrates that the self-supervised film-scoring rules provide strong, learnable signal.

9 Analysis

Domain gap quantification. The drop from GoEmotions F1 (0.682) to direct movie script F1 (0.22) represents a 67.6% relative degradation, quantifying the Reddit-to-script domain shift. Scene texts average 60–120 words, frequently ex-

Scene	Emo.	BPM	Harmonic	Instruments
S99E1D1	mad	68	chrom. aggr.	low_str., woodwinds
S99E1D2	neutral	63	chrom. aggr.	solo_piano
S99E1D3	sad	59	mel. minor	high_str., fr.horn

Table 17: Qualitative music descriptors for *"The Break"*, a Friends-style screenplay. Scene 3's shift to sad + melodic minor + high strings is consistent with standard cinematic breakup scoring conventions.

ceeding MAX_LEN=128, causing truncation of emotionally relevant closing dialogue.

Data size bottleneck. Both models exhibit significant overfitting: SELD reaches train_acc=1.000 by epoch 20 while test loss continues rising (1.4 → 3.2). With only 286–314 training scenes, the model memorises training examples rather than generalising. We estimate a minimum of 500 labelled scenes per genre for stable domain adaptation.

Genre diversity. Adding Avatar and Avengers to the training set significantly reduced the `Tension_Action` deficit present in *500 Days* alone (1 scene → 100 scenes combined), directly producing the large TA F1 gains observed post-adaptation.

Annotation quality. The Gemini pipeline produces `validation_issues` metadata automatically flagging anomalies. In our datasets: 12 scenes had missing `scene_text` (skipped), 2 scenes had `null emotional_core` (excluded), and several had clamped `action_intensity` values. The Avengers dataset introduced additional cinematic core labels (`Awe_Wonder`, `Heroic_Triumph`, etc.) which were mapped to our 4 canonical cores via an extended mapping table.

10 Progress and Timeline

10.1 Completed Objectives

- GoEmotions dataset analysis, label selection, and multi-label classification setup
- SELD architecture: design, training, and threshold tuning (Macro F1 = 0.682)
- BiLSTM-Attn architecture: implementation and training (<0.40 macro F1; confirms end-to-end fine-tuning is necessary)
- Automated 34-field schema induction from 11 screenplays via Gemini; annotation pipeline producing 358 labelled scenes across 3 films (*500 Days of Summer*, *Avatar*, *Avengers: Endgame*)
- Emotion-to-cinematic-core mapping schema (9 GoEmotions labels → 4 cinematic cores)
- Domain adaptation for both architectures; comparative evaluation across all four settings (SELD+DA: 0.432 macro F1; BiLSTM+DA: 0.403)
- Module 1 — Dialogue Transformer Encoder (7.66M params): two-phase training on SuperEmotion then MELD; calibrated test F1 = 31.95%
- Module 2 — SceneEncoderV3 (1.0M params): BiLSTM + 9 multi-task heads; test macro F1 = 55.32%, accuracy 70.15%
- Module 3 — NarrativeLSTM (546K params): causal LSTM + TensionMemoryCell + CharacterMemoryBank; arc-phase test F1 = 52.47%
- Module 4 — MusicPlannerV2 (492K params): 576d shared trunk + 13 output heads; average test F1 = 70.04% across 7 classification heads, zero human music labels required
- Full end-to-end pipeline inference notebook with Mode A (pre-generated prompt lookup) and Mode B (live inference on new screenplays); qualitative validation on custom Friends-style screenplay (*"The Break"*)

10.2 Ongoing

- Expanding annotated script corpus to 10+ films for more robust domain adaptation
- Investigating MAX_LEN = 256/512 to reduce truncation effects on long scenes (>128 tokens)
- Cross-validation for more reliable F1 estimates on small sets

10.3 Planned

- End-to-end cross-domain evaluation: run full pipeline on the 358 annotated movie script scenes, comparing Module 4 output descriptors against Gemini `musical_cue_type` and `tension_level` ground truth
- Human evaluation study with film-music practitioners using a 5-point emotional congruence scale
- MusicGen / Lyria 2 integration: convert Module 4 descriptors to audio and measure CLAP similarity scores
- Ablation studies: TensionMemoryCell contribution, CharacterMemoryBank contribution,

Period	Milestone
Feb 2026	GoEmotions baseline; SELD + BiLSTM-Attn; Gemini annotation pipeline; 358 scenes (✓)
March 2026	Domain adaptation; Modules 1–4 trained; pipeline notebook; mid-report (✓)
April 2026	Expanded corpus; cross-domain eval; human eval; ablations; final report

Table 18: Project timeline. Completed milestones marked (✓).

two-phase M1 training gain, and effect of M3 context window size on M4 quality

11 Conclusion

We presented CineEmotion, a two-part system for context-aware music descriptor generation from movie scripts. In the first part, we established a baseline for scene-level emotion classification: our SELD architecture achieves 0.682 macro F1 on GoEmotions and 0.45 macro F1 after domain adaptation on movie scripts, a 150% relative improvement over the 0.18 direct-inference baseline. These experiments quantify the Reddit-to-script domain gap and identify three failure modes — neutral collapse, subtext-expressed distress, and data scarcity — that directly motivate the proposed full pipeline.

In the second part, we proposed a four-module architecture (Dialogue Transformer Encoder → Scene Encoder → Narrative LSTM → Music Planner) that addresses each identified limitation: explicit speaker modelling, utterance-level scene aggregation, causal arc tracking, and multi-head music descriptor generation. Preliminary results on SuperEmotion and MELD yield 31.95% utterance-level emotion F1 and 70.04% average F1 across music descriptor heads. The primary next step is end-to-end evaluation on the annotated movie script corpus established in this work.

A Annotation Schema

Field	Type	Captures
Narrative		
narrative_stage	cat	Story phase
sense_of_scale	cat	Scene scope
Emotion & Tone		
emotional_core	cat	Dominant mood
emotional_intensity	1–10	Emotional strength
emotional_dramatic_shift	bool	Abrupt mood change
tension_level	1–10	Suspense / dread
moral_ambiguity	1–10	Ethical grey area
humor_tone	cat	Comic register
Conflict		
conflict_nature	cat	Pressure type
conflict_type	cat	Conflict scale
action_intensity	1–10	Physical action
violence_level	0–10	Explicit violence
Character		
character_focus	cat[]	POV focus
character_internal_state	cat	Character emotion
character_transformation	bool	Arc change
relationship_status	cat	Inter-char dynamic
Narrative Device		
foreshadowing_callback	cat	Story callbacks
symbolic_motif	text	Named motif
sense_of_repetition	bool	Pattern recurrence
thematic_elements	cat[]	Scene theme
Cinematic Style		
pacing_intensity	1–10	Editorial pace
visual_pacing_style	cat	Cut rhythm
dialogue_prominence	cat	Speech weight
musical_cue_type	cat	Scoring mode
internal_cue_points	obj[]	Intra-scene shifts
soundscape_elements	cat[]	Audio texture
World & Setting		
scene_setting	cat	Physical space
time_period_aesthetic	cat	Era / world type
cultural_influence	cat	Cultural flavour
reality_distortion	cat	Narrative layer
memory_degradation	cat	Memory reliability
mystical_type	cat	Supernatural kind
spiritual_presence	1–10	Otherworldly feel
technological_prominence	1–10	Tech salience

Table 19: Annotation schema (34 fields, 7 groups). *cat*=categorical; *bool*=boolean; *text*=free text; *cat[]/obj[]*=arrays.

B Code and Resources

All code, annotated datasets, and trained model checkpoints are available in our public GitHub repository:GitHub Repository

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